Ashim Dahal

Undergraduate Researcher in Computer Vision & Multimodal Learning

codeashim@gmail.com | linkedin.com/in/ashimdahal | github.com/ashimdahal

Google Scholar | ashimdahal.github.io

Research Interests

3D/4D Computer Vision, Video Understanding, Vision-Language Models, Embodied Perception, Neural Rendering, Efficient Deep Learning

Education

University of Southern Mississippi Aug 2023 – May 2027 (Expected)

Bachelor of Science in Computer Science; GPA: 3.87/4.0 *Hattiesburg, MS, USA*

– Keystone Honors Scholar and merit-based full-tuition scholarship recipient.

Peer-Reviewed Publications

Journal Articles

[J1] A. **Dahal**, R. Abdelfattah, and N. Rahimi, "Gaussian splatting for real-time scene reconstruction: Methods, systems, and trade-offs," *Pattern Recognition*, vol. 183, p. 115 120, 2027, [**Impact Factor: 9.1**] [arXiv].

[J2] A. **Dahal**, A. Ghimire, S. A. Murad, and N. Rahimi, "Redemption score: A multi-modal evaluation framework for image captioning via distributional, perceptual, and linguistic signal triangulation," *IEEE Transactions on Multimedia*, 2026, Accepted; co-first author, [**Impact Factor: 9.9**] [arXiv].

[J3] S. A. Murad, A. **Dahal**, and N. Rahimi, "Multilingual cyber threat detection in Tweets/X using ML, DL, and LLM: A comparative analysis," *IEEE Transactions on Computational Social Systems*, vol. 13, no. 2, pp. 1758–1772, 2026, [**Impact Factor: 4.9**] [arXiv].

[J4] A. **Dahal**, S. A. Murad, and N. Rahimi, "Heuristic comparison of vision transformers against convolutional neural networks for semantic segmentation on remote sensing imagery," *IEEE Sensors Journal*, vol. 25, no. 10, pp. 17 364–17 373, 2025, [**Impact Factor: 4.5**] [arXiv].

Workshop Papers

[W1] A. **Dahal**, A. Ghimire, S. A. Murad, and N. Rahimi, "POVQA: Preference-optimized video question answering with rationales for data efficiency," in *IEEE/CVF (**CVPRW**)*, Proceedings Track, 2026 [arXiv].

[W2] A. **Dahal**, S. A. Murad, and N. Rahimi, "Embedding shift dissection on CLIP: Effects of augmentations on VLMs' representation learning," in *IEEE/CVF (**CVPRW**)*, Proceedings Track, 2025 [arXiv].

Preprints

[P1] A. **Dahal** and B. Banerjee, *Metasampling: Making frame samplers efficient for long-video question answering*, 2026, Submitted to CVPR [arXiv].

[P2] A. **Dahal**, R. Abdelfattah, and N. Rahimi, *Does it matter which gaussians you pick in 4d gaussian streaming?* 2026, Submitted to CVPR [arXiv].

Research Experience

MCTS-VQA: Monte Carlo Tree Search for Video Evidence Collection [P1] Jan 2026 – Present

Undergraduate Researcher

Advisor: Bikramjit Banerjee

– Developing a Monte Carlo Tree Search framework for identifying and collecting question-relevant frame evidence in long-form video question answering.

– Developed MetaSampling to reduce frame counts by up to 18% while improving accuracy by up to +3.35% [P1].

Anchor Policies for Efficient 4D Gaussian Streaming [J1], [P2] Apr 2025 – Jan 2026

Student Principal Investigator

Advisors: Rabab Abdelfattah and Nick Rahimi

– Developed adaptive anchor-management policies for efficient streaming of dynamic 4D Gaussian representations.

– Investigated anchor insertion, retention, and removal policies to improve memory efficiency while preserving temporal and rendering quality.
– Independently proposed and led the project through a competitive $5,500 DCUR Summer Research Grant.

Cyber Innovations Lab, USM [J2] [J3] [J4] [W1] [W2] Aug 2023 – Present
Undergraduate Research Assistant (fully funded); Advisor: Nick Rahimi *Hattiesburg, MS, USA*
– **POVQA**: Developed a rationale-guided video QA framework that reduced input-token use by 40% while retaining over 95% of baseline accuracy. **[W1]**
– **Redemption Score**: Developed a multimodal image-caption evaluation framework combining distributional, perceptual, and linguistic signals for robust image-text alignment assessment. **[J2]**
– **CLIP Representation Analysis**: Led controlled ablation studies measuring embedding shifts caused by image augmentations in CLIP and related vision-language models. **[W2]**
– **Remote-Sensing Segmentation**: Compared Vision Transformers and convolutional neural networks for semantic segmentation on satellite imagery and developed a weighted loss function for imbalanced classes. **[J4]**
– **NASA EPSCoR Project**: Co-designed the research methodology for a $51,000 NASA EPSCoR-funded project on efficient vision-language models, including objectives, evaluation protocols, and deployment strategies.

Honors & Awards

$1,500 Eagle SPUR Research Grant – Drapeau Center for Undergraduate Research 2026
$500 Best Presentation on Computational Approaches – Drapeau Center for Undergraduate Research 2026
$5,500 Summer Research Grant – Drapeau Center for Undergraduate Research 2025
$51,000 NASA EPSCoR Project – Research Team Member & Technical Proposal Co-Drafter 2025 – 2026
Keystone Honors Scholar – University of Southern Mississippi 2025 – 2027
$500 Hatchery Checkpoint Award 2024
$200 Eagles Write Award – School of Humanities, USM 2024

Leadership & Service

Lead Organizer 2025 – Present
Google Developer Groups on Campus at University of Southern Mississippi *Hattiesburg, MS*
– Organize GDG seminars, a Robo Soccer tournament, and twice-per-semester all-community meetings.
– Represented USM at Google I/O in Sunnyvale, California, in May 2026 as a sponsored delegate.

Research Liaison 2025 – Present
School of Computing Sciences and Computer Engineering Student Ambassadors *Hattiesburg, MS*
– Represent undergraduate research interests and connect prospective students with research opportunities and faculty mentors.

Head of Artificial Intelligence 2024 – 2025
Google Developer Student Club at University of Southern Mississippi *Hattiesburg, MS*
– Delivered the "What Is an Image?" workshop on digital images, color spaces, and image processing to 60+ undergraduate students.
– Led AI/ML workshops and mentored 20+ students on machine-learning projects and research directions.

Teaching & Mentorship Experience

Machine Learning Bootcamp Series 2022 – 2023
Lead Instructor, Data Research Council for Students *Nepal*
– Designed and taught multi-day Python and machine-learning programs covering programming fundamentals, CNNs, PyTorch, and hands-on projects.

Robotics Instructor 2021 – 2022
Durbar High School *Kathmandu, Nepal*
– Designed and taught robotics lessons covering Arduino programming, sensor integration, and mechanical design.

Technical Skills

Programming: Python, C++, CUDA, Bash, SQL, JavaScript
Machine Learning: PyTorch (DDP, FSDP, Lightning), Hugging Face Transformers, Accelerate, scikit-learn, XGBoost
Computer Vision: 3D/4D Gaussian Splatting, NeRFs, Vision Transformers, CLIP, video understanding, image segmentation, multiview stereo
Systems & Tools: SLURM/HPC, Docker, Git/GitHub, Weights & Biases, NumPy, Pandas, OpenCV, Linux, LaTeX